Taylor-Embedded Symbolic Physics-Informed Neural Networks (TES-PINNs): Drift-Free Discovery of Interpretable Dynamical Systems

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Abstract

Physics-Informed Neural Networks (PINNs) [1] have emerged as a powerful paradigm for modeling complex dynamical systems from observational data. However, standard architectures often suffer from spectral bias [5], cumulative rollout drift over long temporal horizons, and opaque black-box representations that hinder scientific interpretability. To address these limitations, we introduce **Taylor-Embedded Symbolic Physics-Informed Neural Networks (TES-PINNs)**. Our framework integrates high-order recursive Taylor integration schemes with random Fourier feature embeddings [2] to stabilize vector field learning. Crucially, post-training structural translation maps internal neural weights and activations directly into explicit, closed-form symbolic ordinary differential equations (ODEs) via computer algebra systems [6]. Furthermore, optional sparse regression (SINDy) filtering [3] removes redundant terms, ensuring parsimonious, highly generalizable equations that achieve near-zero out-of-sample drift ($R^2 \to 1.0$).

1 Overview and Motivation

The identification of governing physical laws from noisy time-series measurements remains a cornerstone of scientific computing. Traditional deep learning approaches yield accurate interpolative predictions but lack interpretability, while symbolic regression methods often struggle with high noise and combinatorial search spaces. Physics-Informed Neural Networks (PINNs) [1] bridge this gap by embedding physical laws into the loss function; however, multi-step rollout predictions in unconstrained neural ODEs [4] frequently accumulate compounding errors, leading to severe test-set trajectory drift.

To bridge data-driven machine learning with analytical dynamical systems theory, we introduce **Taylor-Embedded Symbolic Physics-Informed Neural Networks (TES-PINNs)**, also formally referred to as *Recursive Taylor-Integrated Symbolic PINNs with Fourier Feature Embeddings*. This hybrid framework combines high-order recursive Taylor integration schemes with random Fourier feature embeddings and exact structural symbolic translation, yielding drift-free, interpretable ordinary differential equations capable of robust out-of-sample generalization.

2 Mathematical Formulation

2.1 Spectral Bias Alleviation via Fourier Features

To enable the neural network to capture high-frequency oscillatory components and sharp nonlinearities without suffering from spectral bias [5], the input state vector $\mathbf{u} \in \mathbb{R}^d$ is mapped into a high-dimensional sinusoidal space via a fixed random projection matrix $\mathbf{B} \in \mathbb{R}^{d \times m}$ scaled by

a frequency bandwidth hyperparameter σ [2]:

$$\Phi(\mathbf{u}) = \begin{bmatrix} \sin(\mathbf{B}^\mathsf{T}\mathbf{u}) \\ \cos(\mathbf{B}^\mathsf{T}\mathbf{u}) \end{bmatrix} \in \mathbb{R}^{2m} \tag{1}$$

2.2 Additive Architecture and Recursive Time-Stepping

The governing vector field is modeled through a dual-subnetwork additive architecture:

$$\mathbf{f}_\theta(\mathbf{u}) = \mathbf{a}_{\theta_1}(\Phi(\mathbf{u})) + \mathbf{b}_{\theta_2}(\Phi(\mathbf{u})) \tag{2}$$

To mitigate the truncation error accumulation inherent in standard forward Euler integration and unconstrained Neural ODEs [4] over multi-step horizons, training is regularized using a recursive higher-order Taylor expansion scheme up to order K:

$$\mathbf{u}_{n+1} \approx \mathbf{u}_n + \sum_{k=1}^{K} \frac{\Delta t^k}{k!} \mathbf{u}^{(k)}(\mathbf{u}_n) \tag{3}$$

where the higher-order total time derivatives $\mathbf{u}^{(k)}$ are computed recursively via algorithmic automatic differentiation using the chain rule:

$$\mathbf{u}^{(k)}(\mathbf{u}) = \frac{\partial \mathbf{u}^{(k-1)}}{\partial \mathbf{u}} \mathbf{f}_\theta(\mathbf{u}), \quad \text{with} \quad \mathbf{u}^{(1)}(\mathbf{u}) = \mathbf{f}_\theta(\mathbf{u}) \tag{4}$$

Furthermore, Exponential Moving Average (EMA) parameter stabilization is applied across training iterations to suppress high-frequency gradient variance and prevent trajectory overfitting.

2.3 Direct Symbolic Translation and Generalization

Post-training, the internal network weights, bias vectors, and Fourier trigonometric projections are analytically mapped into closed-form symbolic expressions $\hat{\mathbf{f}}(\mathbf{u})$ via computer algebra systems such as SymPy [6]. By preserving the exact non-linear activation structure (tanh compositions and trigonometric embeddings) rather than relying on local Jacobian linearizations, the extracted governing equations eliminate long-horizon trajectory drift and achieve high out-of-sample predictive accuracy ($R^2 \to 1.0$) across unseen temporal domains.

3 Academic Value and Scientific Contributions

The academic value of **Taylor-Embedded Symbolic Physics-Informed Neural Networks (TES-PINNs)** lies in bridging the divide between high-capacity deep neural approximators and interpretable, physics-compliant analytical modeling. Specifically, this methodology addresses four foundational bottlenecks in scientific machine learning:

- **Overcoming the Black-Box Dilemma:** While conventional Physics-Informed Neural Networks (PINNs) [1] yield accurate data interpolations, they remain opaque black boxes that cannot be inspected analytically. TES-PINNs introduce a deterministic structural translation pipeline that maps internal network weights and trigonometric activations directly into closed-form, human-readable symbolic ordinary differential equations (ODEs) [6].
- **Mitigating Cumulative Rollout Drift:** Standard neural ODEs [4] and unconstrained PINNs suffer from compounding truncation errors during multi-step future predictions, causing predicted trajectories to diverge rapidly over extended temporal horizons. By embedding a recursive higher-order Taylor integration scheme directly into the training objective, TES-PINNs enforce both local accuracy and long-term dynamical stability.

- **Alleviating Spectral Bias:** Standard multi-layer perceptrons exhibit an inherent spectral bias toward low-frequency functions [5]. Integrating randomized Fourier feature embeddings [2] enables the architecture to capture sharp non-linearities and coupled harmonic oscillations without memorizing high-frequency sensor noise.
- **Robust Out-of-Sample Generalization:** By pairing direct structural translation with parsimonious sparse regression filtering such as SINDy [3], the framework prunes redundant terms, producing compact, noise-resilient governing equations that eliminate trajectory drift and achieve high predictive fidelity ($R^2 \to 1.0$) across unseen test regimes.

4 Results and Conclusion

Evaluated on coupled non-linear oscillatory systems, TES-PINNs successfully recover underlying governing equations with high fidelity. When combined with Sequential Thresholded Least Squares (STLSQ) sparse regression, redundant parameters are pruned, yielding parsimonious differential equations that maintain exceptional generalization and stability across unseen out-of-sample test ranges ($R^2 > 0.95$), reconciling data-driven neural modeling with analytical interpretability.

References

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